Problem 10

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Problem 1 *Find a formula for the differential of the following functions.*

(a) $y = 3x^6e^x$.

Explanation. $\dfrac{dy}{dx} = 18x^5e^x + 3x^6e^x$ so

$$\begin{aligned} dy &= \frac{dy}{dx}dx \\ &= \left(18x^5e^x + 3x^6e^x\right) dx. \end{aligned}$$

(b) $z = \ln(1 + t^2)$.

Explanation. $dz = \dfrac{2t}{1 + t^2}dt$

(c) $\theta = \tan^{-1}(r^3)$.

Explanation. $d\theta = \dfrac{3r^2}{1 + r^6}dr$